Quantization and confidence

A calibration-transfer audit of small Qwen3 models

Emmet Surmeli | Independent research project | Measured local deployment study

Abstract

Can a language model retain its existing confidence calibration after weight quantization? We compare transferred BF16 temperature scaling with a scalar temperature refitted at each precision. The controlled intervention is MLX affine weight quantization at 8 and 4 bits, with one pinned checkpoint per model, identical prompts across precisions, and no model training. The primary comparison uses Qwen3-4B on 1,172 held-out ARC-Challenge questions and 400 disjoint calibration examples.

Primary Brier difference	95% paired bootstrap interval	Practical threshold
0.0017	[-0.0039, 0.0083]	0.0100 absolute

The interval supports little practical difference within the prespecified margin in this setup.

Positive differences favor refitting. The interval includes uncertainty from both calibration sampling and test sampling. It is conditional on the selected checkpoints, tasks, and scoring procedure.

Question, baseline, and contribution

The operational baseline is to reuse the temperature already fitted to the BF16 model. Comparing refitting only with raw probabilities would answer an easier question. Secondary analyses examine 8-bit transfer, 50/100/200-example nested calibration subsets, ARC-Easy, and the 1.7B scale replication.

Temperature scaling [1], quantization-induced confidence changes [2], and calibration repair [3] are established. Our contribution is an applied replication and extension: paired transfer evaluation, label-budget sensitivity, and deployment measurements in a reproducible local MLX pipeline. We do not establish that this combination is unprecedented. Preserving BF16 score behavior [4] and improving correctness calibration against labels are different objectives.

What the endpoint measures

For question i and available option k, $p_{ik}(T) = \mathrm{softmax}(z_i/T)_k$. Brier = $\mathrm{mean}_i \, \mathrm{sum}_k \, (p_{ik} - 1[y_i=k])^2$. We report the sum-of-squares convention, with range 0 to 2. The endpoint is Brier(4-bit, transferred) minus Brier(4-bit, refitted).

Brier score measures overall probability quality, not calibration alone. A positive scalar temperature preserves the highest-scoring option, so calibration cannot improve answer accuracy. All probabilities are conditional on the listed choices; this is not a study of free-form hallucinations.

1. Controlled design and prediction changes

The official Qwen3-4B and Qwen3-1.7B checkpoints [6] were converted to BF16, then to affine 8-bit and 4-bit weights with group size 64 in MLX-LM [7]. Activations remain BF16. All runs use batch size one and independent forward passes without a persistent KV cache. The official chat template disables thinking and requests a single option letter. Verified single-token letter logits are normalized over the available options; no question is silently truncated.

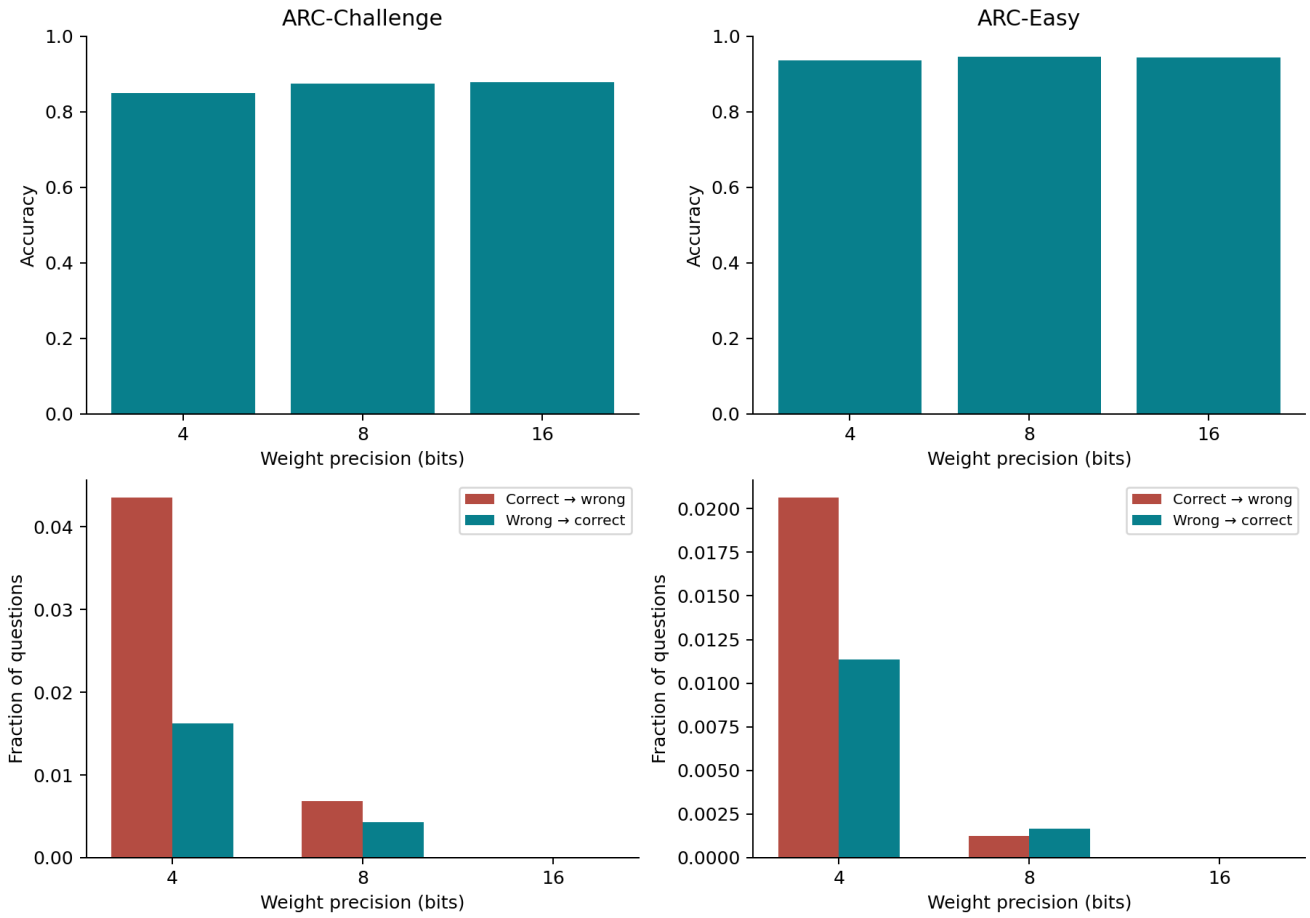


Figure 1. Qwen3-4B accuracy and paired prediction transitions relative to BF16. Counts below retain the question as the unit of observation. On Challenge, 4-bit accuracy drops by 2.73 percentage points for 4B and 7.68 points for 1.7B. Calibration cannot reverse these prediction changes.

Model	Subset	Test n	BF16 acc.	4-bit acc.	Beneficial / harmful
Qwen3-4B	Challenge	1172	0.8771	0.8498	19 / 51
Qwen3-4B	Easy	2376	0.9440	0.9348	27 / 49
Qwen3-1.7B	Challenge	1172	0.7184	0.6416	89 / 179
Qwen3-1.7B	Easy	2376	0.8674	0.7942	96 / 270

Data separation. Each ARC subset [5] contributes 100 training questions to the pilot, 400 different training questions to calibration, and 100 validation questions to pre-test checks. Global screening excluded 20 exact-content or cross-split stem matches; all 1,172 Challenge and 2,376 Easy test questions remain. The 200-question permutation subset per task is chosen deterministically. Both models passed the pilot. An initial freeze preceded test scoring; a data-only stem-overlap amendment preceded all held-out performance review. Calibration was refitted and frozen anew; unchanged raw scores retain their original provenance. See PROTOCOL-AMENDMENT.md.

2. Calibration transfer on the held-out test

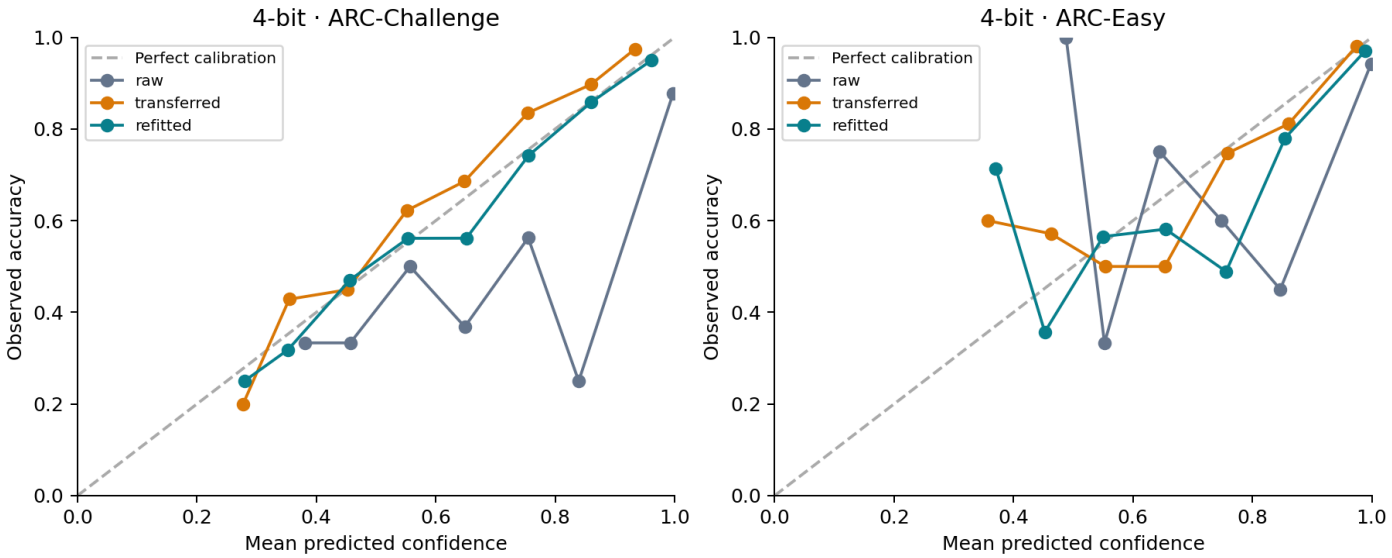


Figure 2. Qwen3-4B, 4-bit reliability diagrams using 10 fixed equal-width confidence bins. Empty bins are omitted. These descriptive curves can be noisy in sparsely populated bins; the primary inference uses paired Brier differences rather than ECE.

Subset	Treatment	Temperature	Brier	NLL	ECE
Challenge	raw	1.0000	0.2758	1.4804	0.1290
Challenge	transferred	6.6971	0.2252	0.4623	0.0444
Challenge	refitted	5.2601	0.2234	0.4491	0.0157
Challenge	uniform	-	0.7498	1.3859	-
Easy	raw	1.0000	0.1244	0.7092	0.0610
Easy	transferred	4.9982	0.1030	0.2185	0.0151
Easy	refitted	3.9016	0.1059	0.2280	0.0277
Easy	uniform	-	0.7498	1.3858	-

The primary Brier score changes from **0.2252** after transfer to **0.2234** after refitting. The absolute improvement is **0.0017**, with a 95% interval of **[-0.0039, 0.0083]**. The prespecified operational margin is 0.01, rather than a threshold selected after seeing the data.

Uncertainty and decision rule

We use 2,000 paired bootstrap replicates. Within each replicate, calibration questions are sampled with replacement using the same indices for BF16 and quantized scores; both temperatures are refitted by minimizing NLL over log-temperature, bounded to T in [0.05,20]. Test questions are resampled independently, retaining pairing between treatments. A useful positive headline requires a positive interval and an estimated improvement of at least 0.01. An interval within [-0.01,0.01] supports little practical difference only in this setup. Crossing zero alone does not establish equivalence.

3. Label budgets and scale replication

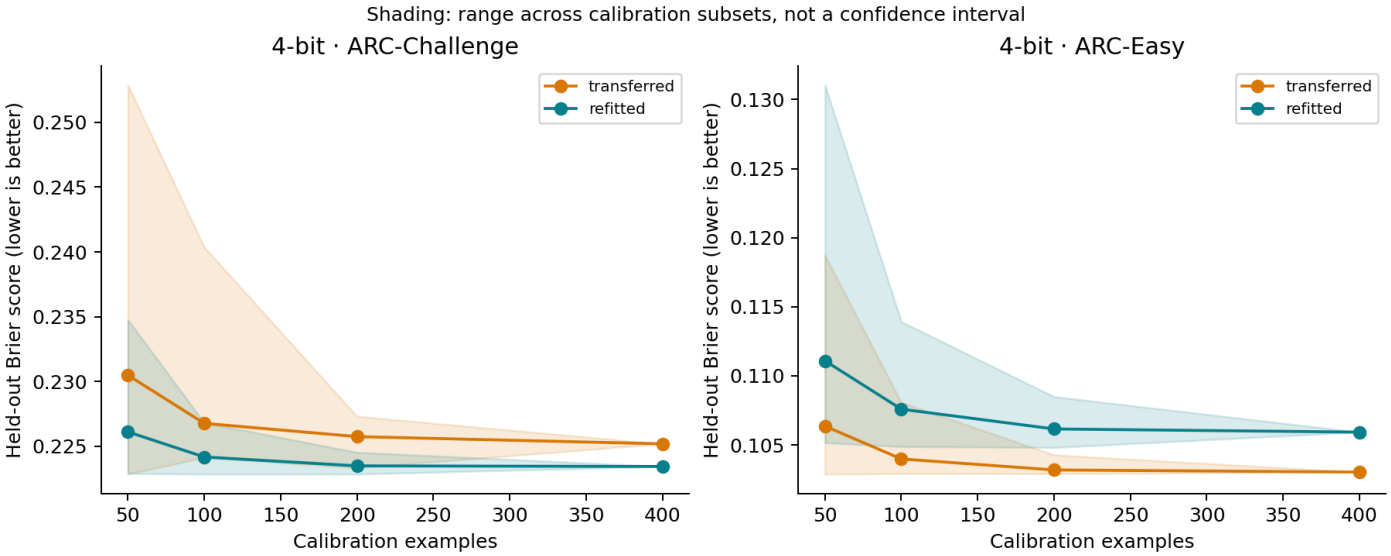


Figure 3. Held-out Brier score versus calibration sample count for Qwen3-4B at 4 bits. Both transferred and refitted temperatures use the same subset IDs and size. Shading is the range across ten seeded nested calibration subsets, not a confidence interval. At 400 all subsets coincide.

Subset	Calibration n	Mean refitted Brier	Range over subsets
Challenge	50	0.2261	0.2229 - 0.2348
Challenge	100	0.2242	0.2228 - 0.2268
Challenge	200	0.2235	0.2228 - 0.2245
Challenge	400	0.2234	0.2234 - 0.2234
Easy	50	0.1111	0.1052 - 0.1311
Easy	100	0.1076	0.1049 - 0.1139
Easy	200	0.1062	0.1048 - 0.1085
Easy	400	0.1059	0.1059 - 0.1059

The subsets overlap and share one test set. They measure sensitivity to label selection; they are not ten independent replications. The 400-example fit is a reference, not a proven optimum. The 50-example curves are more variable; some primary-task fits are over 0.01 worse than the 400-example refit. The 100/200-example curves are tighter in these subsets, without establishing a universal budget. 1 of 744 sample-curve rows reach a temperature boundary (rows share fits and are not independent). Boundary flags and exact temperatures are saved in sample-size.csv.

Secondary 400-example transfer differences

Model	Subset	8-bit difference	4-bit difference
Qwen3-4B	Challenge	0.0002	0.0017
Qwen3-4B	Easy	-0.0003	-0.0029
Qwen3-1.7B	Challenge	-0.0011	0.0026
Qwen3-1.7B	Easy	-0.0004	0.0005

Positive differences favor refitting. Only the prespecified primary comparison carries the primary bootstrap decision. These secondary comparisons are exploratory; no family of significance claims is made. ARC-Easy and ARC-Challenge are related tasks, and the models share a family.

4. Selective answering and deployment cost

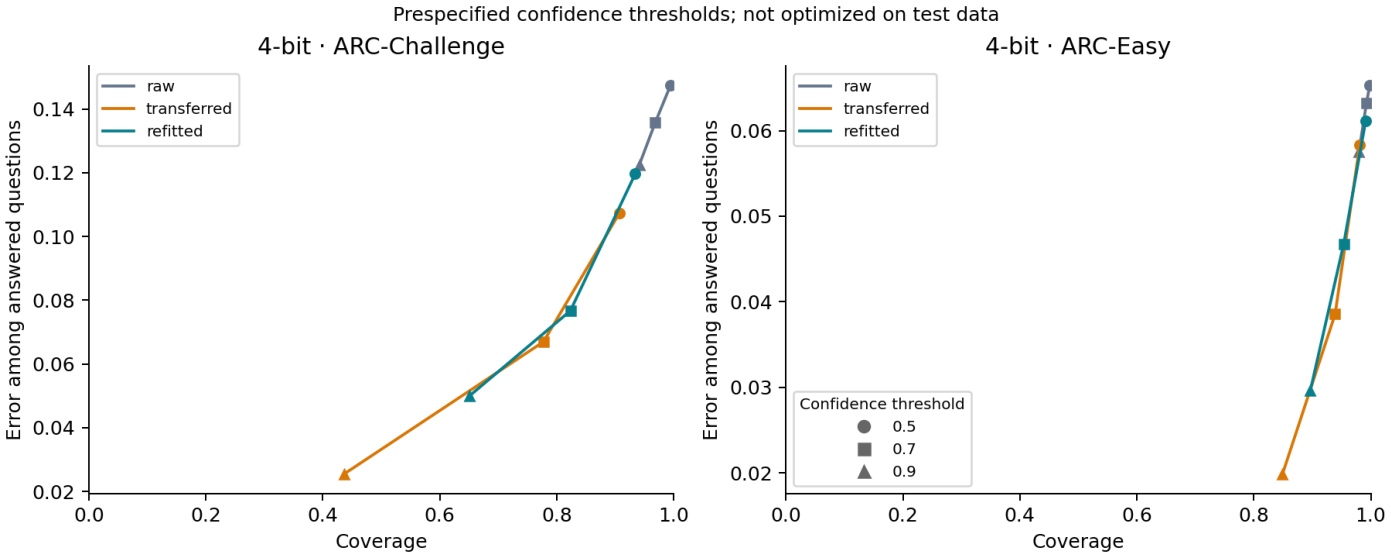


Figure 4. Qwen3-4B, 4-bit error among answered questions versus coverage at fixed confidence thresholds 0.5, 0.7, and 0.9. Connected points are descriptive, not a threshold optimized on the test set. Counts and Wilson error intervals are retained in results.json; small groups are flagged.

Model	Bits	Serialized GB	Peak MLX GB	Mean sec/question
Qwen3-4B	BF16	8.056	8.272	0.1353
Qwen3-4B	8	4.286	4.589	0.2081
Qwen3-4B	4	2.275	2.702	0.2289
Qwen3-1.7B	BF16	3.453	3.648	0.0759
Qwen3-1.7B	8	1.840	2.218	0.1201
Qwen3-1.7B	4	0.980	1.416	0.1072

Hardware: Apple M5 with 24 GB unified memory. Values are measured on this host, not hardware-independent compute estimates. Serialized size includes model files and quantization metadata. Peak memory is MLX allocation, not total resident system memory. Synchronized per-question latency excludes cold loading, tokenization, and warm-up. Models run sequentially; run order, temperature, and other host activity may affect latency. Quantized kernels need not be faster than BF16.

Interpreting confidence thresholds

Temperature scaling changes which questions pass a fixed confidence threshold while preserving each question's answer. A low observed error on a tiny answered subset is not evidence of reliable deployment. The error-rate intervals are descriptive binomial intervals conditional on the fitted treatment and threshold, not simultaneous risk-control guarantees. No threshold was tuned using test labels.

Additional spending: $0. Both pilot runtime projections passed the 24-hour cap. Model weights are kept outside Git; the repository includes a portable archive of scores and provenance.

5. Robustness, limitations, and reproducibility

Model	Subset	Pairs	Benefit: orig. / perm.	Accuracy change
Qwen3-4B	Challenge	200	0.0071 / 0.0008	-0.0200
Qwen3-4B	Easy	200	0.0006 / -0.0044	-0.0050
Qwen3-1.7B	Challenge	200	0.0017 / 0.0047	-0.0350
Qwen3-1.7B	Easy	200	0.0004 / 0.0001	0.0300

One prespecified nonidentity option permutation per question. Benefit is transferred minus refitted Brier on each version; positive favors refitting. Accuracy change is permuted minus original on the same 200-question subset. Frozen temperatures are reused. These descriptive paired checks are not 400 independent examples. Full precision/treatment results are in robustness.csv.

Error inspection and scope

20 cases were selected by a fixed rule: up to five each of unchanged-correct, unchanged-wrong, beneficial flips, and harmful flips in the primary comparison. The supporting case review records questions, original labels, canonical mappings, selected answers, and probabilities. It flags a water-cycle item where both evaporation and transpiration are valid, and a clear ion-charge error after quantization. Mappings match and labels are unchanged. No reasoning traces establish why a flip occurred.

Benchmark contamination remains unresolved. Paired comparisons control many implementation differences but cannot remove pretraining exposure. Exact normalized-content deduplication does not detect all semantic duplicates. Two related ARC subsets and two sizes of one family do not establish generality. Results concern this MLX affine quantizer, not GPTQ, AWQ, all 4-bit models, or open-ended confidence. A uniform baseline and agreement with BF16 do not make BF16 an oracle.

Reproducibility and next question

The repository records split IDs, immutable source revisions, prompt-token hashes, option scores, quantized-module inventories, calibration fits, environment versions, and question-level cached output. The analysis archive restores without model weights; all four figures and tables regenerate from saved scores. Automated checks cover hand-calculated metrics, label permutations, data separation, paired IDs, immutable resume behavior, and archive round trips. Implementation was AI-assisted; interpretation and ownership require understanding the controls, held-out design, and claim limits.

A focused follow-up is to preregister the same transfer test on a different model family and a task outside ARC, while keeping the score definition and practical margin fixed. This would test whether the observed transfer behavior survives a meaningful change of setting.

References and source artifacts

[1] Guo et al. (2017). On Calibration of Modern Neural Networks. ICML.

[2] Proskurina et al. (2024). When Quantization Affects Confidence of Large Language Models? NAACL Findings.

[3] Zhong et al. (2025). Quantized Can Still Be Calibrated. ACL.

[4] Yang et al. (2026). Target-Aware Calibration Data Selection for Preserving Uncertainty in Quantized Language Models. arXiv:2608.21019.

[5] Allen Institute for AI. AI2 ARC dataset, pinned revision 210d026faf99.

[6] Qwen. Qwen3-4B / 1.7B; revisions 1cfa9a720891 / 70d244cc86cc.

[7] ML Explore. MLX-LM 0.31.3; MLX 0.32.3.

Repository: github.com/EmmetSurmeli/quantization-audit